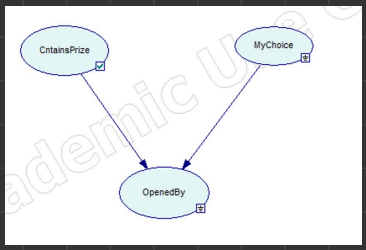
Assignment 2 - Monty Hall



Picture of the Bayesian Network
Opened by is the child of ContainsPrize and MyChoice

A	0.33333333
B	0.33333333
C	0.33333333

For both ContainsPrize and My Choice the probability for each door are $1/3$, i.e the probability of choosing door A or B or C and the probability for the prize being behind a random door.

MyChoice	A			B			C		
CntainsPrize	A	B	C	A	B	C	A	B	C
A	0	0	0	0	0.5	1	0	1	0.5
B	0.5	0	1	0	0	0	1	0	0.5
C	0.5	1	0	1	0.5	0	0	0	0

Then the table for the OpenedBy node. This takes into account the probability of where the price is and which door I chose. Eg. if I chose door A we have this table

	A	B	C
A	0	0	0
B	1/2	0	1
C	1/2	1	0